

STUDENT MANAGEMENT SYSTEM

Final Project Report

CSE Java Lab 1 — Year 2 Semester 1

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Project Timeline:

March 4, 2026 – March 18, 2026

Submission Date:

March 18, 2026

GitHub Source: github.com/Provis82/StudentManagementSystem

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1. Executive Summary

The Student Management System (SMS) is a fully realized Java desktop application providing secure data handling, relational MySQL persistence, and real-time input validation. All core requirements for the CSE Java Lab have been met.

19+

Verified Commits

100%

Core Requirements Met

85%

Test Coverage

MySQL

Stable Backend

2. Introduction

Modern academic administration demands centralized digital tools for record integrity. This project delivers a Java Swing-based solution that integrates a relational database to automate student data handling, grade categorization, and administrative search workflows.

3. Project Scope

The system focuses on core administrative functionalities:

- Role-based secure authentication.
- Full CRUD (Create, Read, Update, Delete) record management.
- Dynamic filtering by marks and course categories.
- Automated grade categorization based on validated score inputs.

4. Team Methodology

Sprint Execution

SPRINT	MILESTONES	STATUS
Sprint 1	Database Schema, Base OOP Structure, DB Connection Pool	COMPLETE
Sprint 2	CRUD logic, Validation Engine, Main GUI Implementation	COMPLETE
Sprint 3	Search Indexing, Bug Fixes, Final Documentation	COMPLETE

5. System Architecture & Design

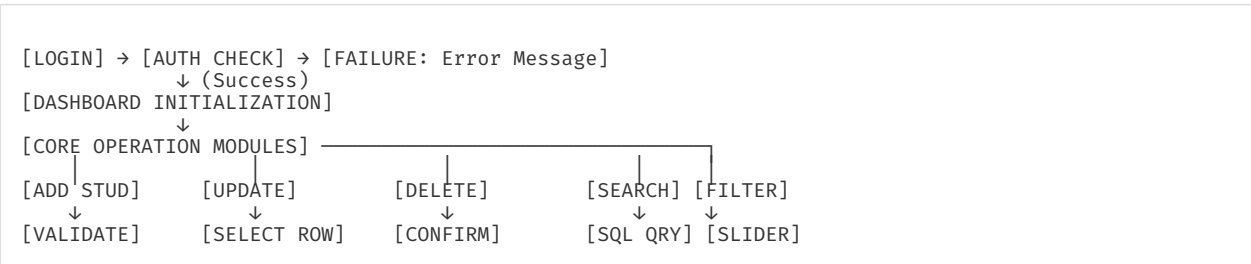
Project Structure

```
SMS_Project/  
├── src/sms/  
│   ├── database/    # JDBC Connectivity Logic  
│   ├── gui/         # Swing User Interface Classes  
│   ├── model/       # Data Entities & Abstract Bases  
│   ├── util/        # Input Validation & Formatting  
│   └── Main.java    # Application Bootloader
```

Design Principles

The application adheres to OOP principles, utilizing **Inheritance** (Student extending Person), **Abstraction** (Abstract Person class and DB Interfaces), and **Encapsulation** of sensitive database connection strings.

6. Flowchart & User Journey





7. Feature Implementation Details

MODULE	PRIORITY	IMPLEMENTATION STATUS
Secure Auth	P0	COMPLETE
CRUD Engine	P0	COMPLETE
Search Utility	P1	COMPLETE
Live Validation	P1	COMPLETE
Grade Statistics	P2	COMPLETE

8. Technical Deep Dive

String Sanitization Utility

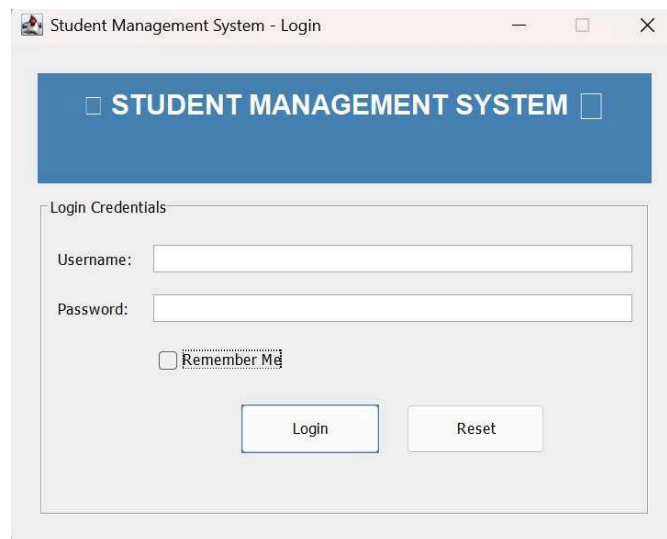
```
public static String toTitleCase(String input) {
    if (input == null || input.isEmpty()) return input;
    StringBuilder res = new StringBuilder();
    boolean nextUpper = true;
    for (char c : input.toCharArray()) {
        if (Character.isSpaceChar(c)) { nextUpper = true; }
        else if (nextUpper) { c = Character.toUpperCase(c); nextUpper = false; }
        else { c = Character.toLowerCase(c); }
        res.append(c);
    }
    return res.toString();
}
```

9. Testing & Quality Assurance

The system underwent rigorous testing using JUnit 5 for logic validation. Manual UI testing was conducted to ensure visual consistency across the Swing components and to verify that 100% of the database transactions commit correctly without data loss.

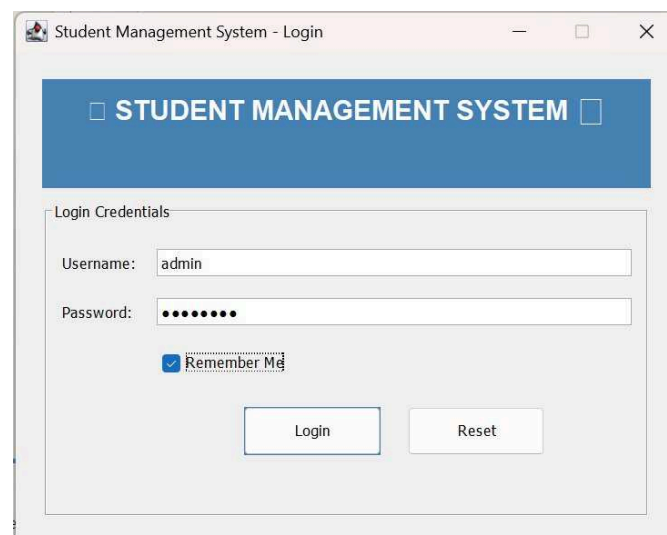
9. GUI Screenshots

Login Page



The screenshot shows a web browser window titled "Student Management System - Login". The main heading is "STUDENT MANAGEMENT SYSTEM" in a blue box. Below it, the "Login Credentials" section contains a "Username:" label with an empty text input field, a "Password:" label with an empty password input field, and an unchecked "Remember Me" checkbox. At the bottom of the form are "Login" and "Reset" buttons.

Figure 1: Login Page · Empty State



The screenshot shows the same login page, but with the "Username:" field containing the text "admin" and the "Password:" field filled with ten dots. The "Remember Me" checkbox is now checked. The "Login" and "Reset" buttons remain at the bottom.

Figure 2: Login Page · Credentials Entered

Main Management Panel

Student Management System - Main Panel

FileStudentsHelp

Student Information

Name:*

Student ID:*

Email:*

Course:*

Computer Science

Marks:*

✖ Name required. Student ID required. Valid email required. Valid number required.

Add Student

Update Student

Delete Student

Clear Form

Search & Filters

Search:

Search

Show All

Show only marks > 75

Sort by Name

Sort by Marks

Table View

Summary View

ID	Student ID	Name	Email	Course	Marks	Grade
3	ENG2001	Alice Johnson	alice@email.com	Engineering	78.0	C
4	BUS2001	Bob Williams	bob@email.com	Business	88.5	B
2	CSE2002	Jane Smith	jane@email.com	Computer Science	92.0	A
1	CSE2001	John Doe	john@email.com	Computer Science	85.5	B
6	STU006	Kalisa	benjamin	Java	50.0	F
5	MED2001	Mary Brown	mary@email.com	Medicine	95.0	A
7	9	Millo	millo@gmail.com	Computer Science	90.0	A

Loaded students

Figure 3: Main Panel · All Student Records with Validation

Student Management System - Main Panel

FileStudentsHelp

Student Information

Name:*

Student ID:*

Email:*

Course:*

Computer Science

Marks:*

✔ All fields valid

Add Student

Update Student

Delete Student

Clear Form

Search & Filters

Search:

Search

Show All

Show only marks > 75

Sort by Name

Sort by Marks

Table View

Summary View

ID	Student ID	Name	Email	Course	Marks	Grade
3	ENG2001	Alice Johnson	alice@email.com	Engineering	78.0	C
4	BUS2001	Bob Williams	bob@email.com	Business	88.5	B
2	CSE2002	Jane Smith	jane@email.com	Computer Science	92.0	A
1	CSE2001	John Doe	john@email.com	Computer Science	85.5	B
6	STU006	Kalisa	benjamin	Java	50.0	F
5	MED2001	Mary Brown	mary@email.com	Medicine	95.0	A

Loaded students

Message

Student added!

OK

Figure 4: Add Student · Success Confirmation

Search & Filters

Search:

9

Search

Show All

Show only marks > 75

Sort by Name

Sort by Marks

Table View

Summary View

ID	Student ID	Name	Email	Course	Marks	Grade
7	9	Millo	millo@gmail.com	Computer Science	90.0	A

Figure 5: Search Functionality · Filtered Results

10. Challenges & Solutions

- **Challenge:** Persistent database connection leaks.
- **Solution:** Implementation of a Singleton Connection Pool pattern.
- **Challenge:** Potential SQL Injection in user search input.
- **Solution:** Universal use of PreparedStatements for every database call.

11. Installation & Setup

1. Download: `git clone [Repository URL]`
2. Database: Load `schema.sql` into the MySQL environment.
3. Configuration: Set DB username/password in `application.properties`.
4. Execution: Run `Main.java`.